

MATH 227C: Mathematical Biology

Homework 5

Karthik Desingu

May 24, 2026

Problem 1

Consider the predator-prey model with three reactions that we defined in class: $Y_1 \rightarrow 2Y_1$, $Y_1 + Y_2 \rightarrow 2Y_2$, $Y_2 \rightarrow \phi$, with stochastic rate constants $c_1 = 1$, $c_2 = 0.005$, $c_3 = 0.6$.

a) Write code to simulate this system stochastically, with initial condition of 50 prey and 100 predators. Write the time course both as a function of time and on a phase plot of Y_1 vs Y_2 .

We consider a stochastic predator-prey model with the following three reactions:



where Y_1 is the prey population and Y_2 is the predator population. The stochastic rate constants are $c_1 = 1$, $c_2 = 0.005$, and $c_3 = 0.6$. We simulate the system starting with an initial condition of $Y_1(0) = 50$ and $Y_2(0) = 100$.

Implementation Logic

The system is simulated using the exact Stochastic Simulation Algorithm (SSA), commonly known as the Gillespie Algorithm. At any given state (Y_1, Y_2) , the propensities for the three reactions are calculated as:

$$\begin{aligned} a_1 &= c_1 Y_1 \\ a_2 &= c_2 Y_1 Y_2 \\ a_3 &= c_3 Y_2 \end{aligned}$$

The total propensity is $a_0 = a_1 + a_2 + a_3$. The time until the next reaction, τ , is drawn from an exponential distribution with parameter a_0 , such that $\tau = \frac{1}{a_0} \ln(1/r_1)$, where $r_1 \sim U(0,1)$. The specific reaction that occurs is chosen proportionally to its propensity using a second random number $r_2 \sim U(0,1)$.

The core logic of this implementation is provided in Listing 1.

```

1 while t < t_max and Y1 > 0 and Y2 > 0:
2     # 1. Calculate propensities
3     a1 = c1 * Y1
4     a2 = c2 * Y1 * Y2
5     a3 = c3 * Y2
6     a0 = a1 + a2 + a3
7
8     if a0 == 0: break
9
10    # 2. Draw random numbers
11    r1 = random.random()
12    r2 = random.random()
13
14    # 3. Update time step
15    tau = (1.0 / a0) * np.log(1.0 / r1)
16    t += tau
17
18    # 4. Determine and execute the next reaction
19    if r2 < a1 / a0:
20        Y1 += 1                # Reaction 1
21    elif r2 < (a1 + a2) / a0:
22        Y1 -= 1                # Reaction 2
23        Y2 += 1
24    else:
25        Y2 -= 1                # Reaction 3

```

Listing 1: Core implementation of the Gillespie Algorithm loop

Results

The dynamic behavior of the populations over time is shown in Figure 1. The simulation captures the inherent stochasticity of the system, demonstrating the characteristic out-of-phase oscillations associated with predator-prey dynamics, where an increase in prey is followed by an increase in predators, leading to a subsequent prey collapse.

Figure 2 visualizes the same trajectory mapped onto a Y_1 vs Y_2 phase plane. Unlike deterministic ODE models which often settle into perfect limit cycles, the stochastic trajectories demonstrate noisy spiral pathways, illustrating how intrinsic noise affects the amplitude and period of the population cycles.

b) Simulate the same system deterministically and plot the same graphs in order to compare their behavior.

To compare the stochastic behavior to a macroscopic limit, we formulate the deterministic Ordinary Differential Equations (ODEs) using mass-action

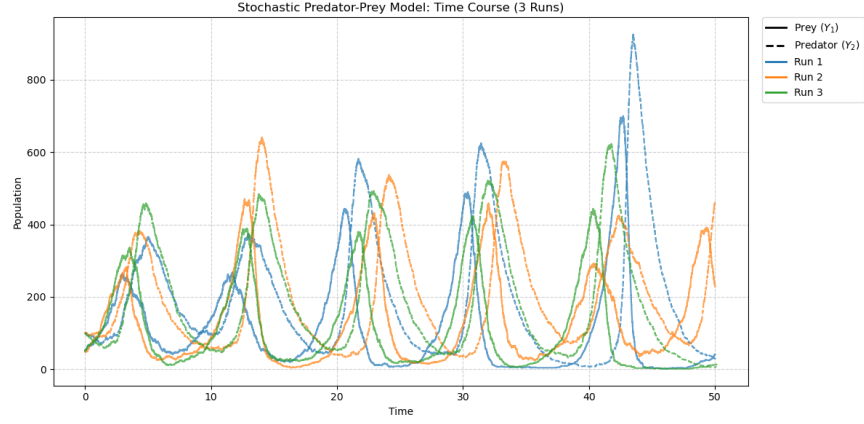


Figure 1: Time course of the predator (Y_2) and prey (Y_1) populations generated via the Gillespie algorithm. Initial conditions: $Y_1 = 50$, $Y_2 = 100$.

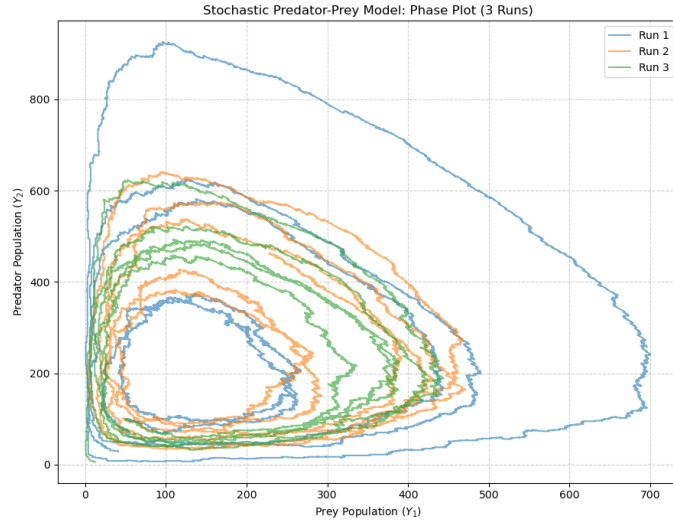


Figure 2: Phase plot of Y_1 versus Y_2 . The noisy orbital paths reflect the stochastic nature of the discrete chemical reactions.

kinetics. Based on the given reaction system, the rates of change for the prey (Y_1) and predator (Y_2) populations are given by:

$$\begin{aligned}\frac{dY_1}{dt} &= c_1 Y_1 - c_2 Y_1 Y_2 \\ \frac{dY_2}{dt} &= c_2 Y_1 Y_2 - c_3 Y_2\end{aligned}$$

This represents the classic Lotka-Volterra equations. The same parameters ($c_1 = 1$, $c_2 = 0.005$, $c_3 = 0.6$) and initial conditions ($Y_1(0) = 50$, $Y_2(0) = 100$) are used.

Implementation Logic

The deterministic system is solved numerically using the `odeint` function from the `scipy.integrate` library, which relies on LSODA for solving standard initial value problems. The core definition of the system is shown in Listing 2.

```

1 def deterministic_predator_preY(Y, t, c1=1.0, c2=0.005, c3=0.6):
2     Y1, Y2 = Y
3
4     # Mass-action kinetics
5     dY1_dt = c1 * Y1 - c2 * Y1 * Y2
6     dY2_dt = c2 * Y1 * Y2 - c3 * Y2
7
8     return [dY1_dt, dY2_dt]
9
10 # Solve the ODEs over t = [0, 50]
11 solution = odeint(deterministic_predator_preY, [50, 100], t)
```

Listing 2: Core implementation of the deterministic ODEs

The deterministic time course (Figure 3) exhibits perfectly smooth, periodic oscillations. In contrast to the stochastic simulation (Figure 1), there is no variation in the amplitude or period of the population cycles. The deterministic model assumes continuous population concentrations, ignoring the discrete nature of individual organisms.

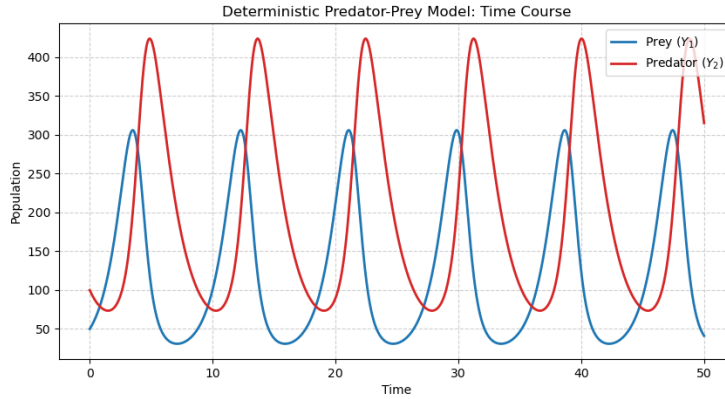


Figure 3: Deterministic time course of the predator and prey populations, exhibiting perfect periodicity without noise.

The difference is most pronounced in the phase plane. The deterministic phase plot (Figure 4) forms a single, perfectly closed limit cycle. Conversely, the

stochastic phase plot (Figure 2) showed erratic, spiral-like orbits. This indicates that while the deterministic ODEs accurately capture the *average* cyclical behavior, they fail to capture the intrinsic noise (demographic stochasticity). In small populations, this stochastic noise can push trajectories away from the deterministic limit cycle, potentially even leading to the random extinction of one species—a phenomenon impossible to predict using only the deterministic model.

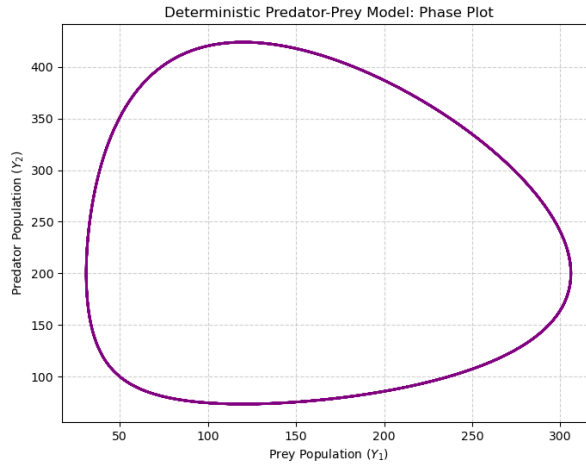


Figure 4: Deterministic phase plot of Y_1 versus Y_2 , forming a perfectly closed loop characteristic of conservative Lotka-Volterra dynamics.

c) Consider lowering the number of predators and prey to the point that the long term behavior of the solution can be qualitatively different from that of the deterministic system. Explain intuitively how this can take place in terms of the predators and prey.

Demographic Stochasticity and Extinction

When the initial populations of predators and prey are significantly reduced (e.g., $Y_1 = 2$ and $Y_2 = 2$), the long-term behavior of the stochastic system diverges entirely from the deterministic model due to demographic stochasticity. In a deterministic framework, population sizes are continuous variables; they can drop to infinitesimally small fractions (e.g., 0.001 predators) and still mathematically recover to continue the cycle. In contrast, the stochastic model treats populations as discrete, whole integers. If a population experiences a random sequence of adverse events and drops to zero, it cannot recover. Zero acts as an absorbing boundary.

Biological Intuition of Absorbing States

Intuitively, in a very small population, random chance plays a disproportionately large role. A few consecutive predator deaths before they manage to find and consume prey, or a rapid succession of predation events, can easily wipe out a small population. This leads to two possible, qualitatively different long-term scenarios that cannot occur in the deterministic ODEs:

- **Predator Extinction:** If the few remaining predators die off by chance ($Y_2 = 0$), predation ceases completely. The prey species, completely unchecked, will undergo unbounded exponential growth driven solely by the reproduction reaction ($Y_1 \xrightarrow{c_1} 2Y_1$).
- **Prey Extinction:** If the predators consume all the remaining prey ($Y_1 = 0$), the predator population loses its only food source. Without prey to facilitate predator reproduction, the predator population will inevitably decay to zero through the death reaction ($Y_2 \xrightarrow{c_3} \emptyset$), resulting in a completely dead system where both species are extinct.

Both of these cases are shown in the runs in Figure 5.

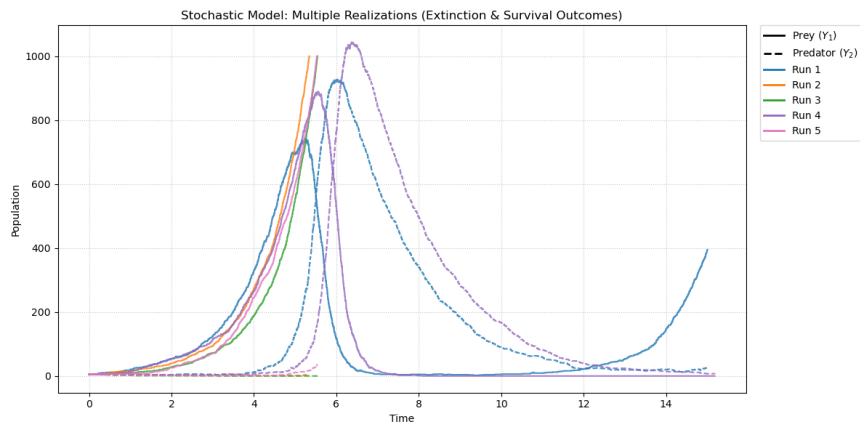


Figure 5: Deterministic time course of the reaction $A + B \rightleftharpoons C$. Starting from initial concentrations $[A] = 10$, $[B] = 5$, $[C] = 0$, the system successfully converges to a steady-state chemical equilibrium.

Divergence from Deterministic Behavior

In the deterministic Lotka-Volterra model, the system is fundamentally conservative, meaning the trajectories form perfectly closed orbits forever. It does not contain absorbing states. Therefore, the deterministic model guarantees infinite

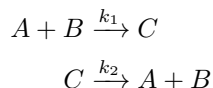
cyclic persistence of both species regardless of how small the numbers get. By lowering the population numbers in the stochastic model, we reveal that these cyclic trajectories are actually transient; given enough time at low numbers, the intrinsic noise of discrete biological events guarantees that the cyclic behavior will permanently collapse into one of the extinction states.

Problem 2

Consider the reaction network $A + B \rightarrow C$, $C \rightarrow A + B$ discussed in class.

a) Simulate this system deterministically, by setting fixed parameter values, and plot the concentration of A, B, C over time.

We consider a reversible reaction network where species A and B associate to form C, and C dissociates back into A and B:



Using the principles of mass-action kinetics, the macroscopic deterministic behavior of the system is governed by the following system of coupled Ordinary Differential Equations (ODEs):

$$\begin{aligned} \frac{d[A]}{dt} &= -k_1[A][B] + k_2[C] \\ \frac{d[B]}{dt} &= -k_1[A][B] + k_2[C] \\ \frac{d[C]}{dt} &= k_1[A][B] - k_2[C] \end{aligned}$$

where brackets denote the concentration of the respective species. Notably, the conservation of mass dictates that the quantities $[A] + [C]$ and $[B] + [C]$ remain constant throughout the reaction.

The resulting time course is presented in Figure 6. As the reaction progresses, the concentrations asymptotically approach a chemical equilibrium. At this steady state, the rate of the forward reaction exactly balances the rate of the backward reaction ($k_1[A][B] = k_2[C]$), yielding a net zero rate of change for all species ($\frac{d[A]}{dt} = \frac{d[B]}{dt} = \frac{d[C]}{dt} = 0$).

b) For the same parameter values as in (a), simulate this system stochastically, and compare the two solutions.

We implement the stochastic analog of the reversible reaction network using the Gillespie algorithm. The reaction propensities are defined as $a_1 = k_1AB$ for the forward reaction and $a_2 = k_2C$ for the backward reaction. The core loop computing these discrete state changes is provided in Listing 3.

```
1 while t < t_max:
```

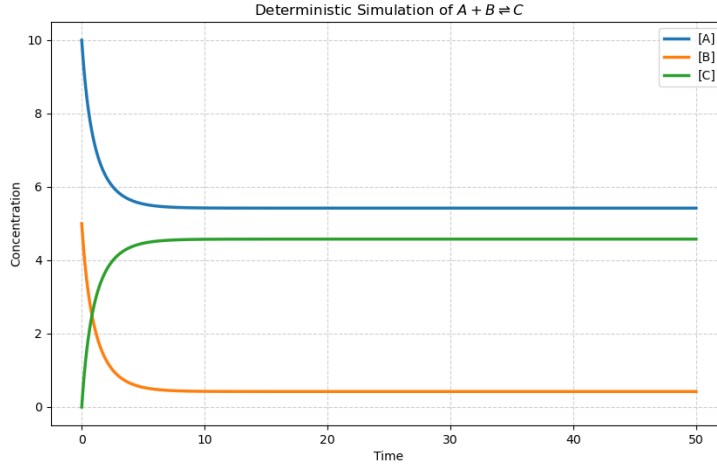


Figure 6: Deterministic time course of the reaction $A + B \rightleftharpoons C$. Starting from initial concentrations $[A] = 10, [B] = 5, [C] = 0$, the system successfully converges to a steady-state chemical equilibrium.

```

2      a1 = k1 * A * B # Propensity of A + B -> C
3      a2 = k2 * C     # Propensity of C -> A + B
4      a0 = a1 + a2
5
6      tau = (1.0 / a0) * np.log(1.0 / random.random())
7      t += tau
8
9      if random.random() < a1 / a0:
10         A -= 1; B -= 1; C += 1 # Forward reaction occurs
11     else:
12         A += 1; B += 1; C -= 1 # Backward reaction occurs

```

Listing 3: Stochastic implementation of the reversible reaction

Comparison of Solutions

Figure 7 overlays 3 stochastic realizations on top of the deterministic ODE solution. While the stochastic trajectory generally follows the mean deterministic path toward equilibrium, it exhibits noticeable discrete fluctuations.

Because the initial particle numbers are very small ($A_0 = 10, B_0 = 5$), the intrinsic noise of the system is highly pronounced. The deterministic model predicts smooth, fractional steady-state concentrations, whereas the stochastic model demonstrates "equilibrium noise"—continuous, integer-level state switching around the equilibrium point as the forward and backward reactions continue to fire dynamically.

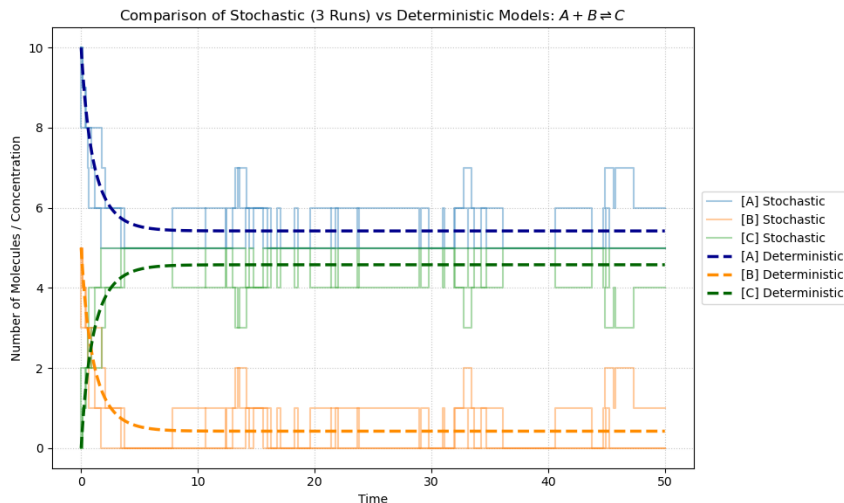


Figure 7: Overlay of the deterministic solution (dashed lines) and a single stochastic realization (solid steps). The stochastic model fluctuates discretely around the deterministic steady-state mean.

c) Extend this comparison to increasingly high values of the initial conditions, simulated over a fixed time interval $[0, T]$. Do you see a convergence between the stochastic and the deterministic solutions?

To investigate the convergence of the stochastic solution to the deterministic one, we systematically scale up the system size. Let V act as a scaling factor representing the system's "volume." To maintain consistent macroscopic dynamics, the initial molecule numbers are scaled linearly with V (e.g., $A_0(V) = 10V$, $B_0(V) = 5V$). Accordingly, to preserve the deterministic rate $k_1[A][B]$, the stochastic bimolecular reaction propensity must be inversely scaled by the volume: $c_1 = k_1/V$. The unimolecular rate constant remains $c_2 = k_2$.

By running the Gillespie algorithm at increasingly high initial condition scales and converting the resulting particle counts back into concentrations (dividing by V), we can directly overlay them on the scale-invariant deterministic solution.

Observation of Convergence

The results are displayed across the six subplots in Figure 8. For small system sizes ($V = 1$, corresponding to $A_0 = 10, B_0 = 5$), the stochastic fluctuations are massive relative to the total population. The step-like nature of individual reaction events is highly visible, and the system deviates significantly from the continuous deterministic curve.

As the scaling factor V increases, we observe a clear and definitive convergence. At $V = 500$ ($A_0 = 5000, B_0 = 2500$), the stochastic solution tracks the deterministic differential equations almost perfectly. The relative amplitude of the noise decreases proportional to $1/\sqrt{V}$, a phenomenon formalized mathematically by Kurtz's limit theorem and the Law of Large Numbers.

Thus, we conclusively see that as the number of reacting molecules approaches the thermodynamic limit, the intrinsic demographic stochasticity "averages out," and the deterministic ODEs become a highly accurate representation of the system's true dynamics.

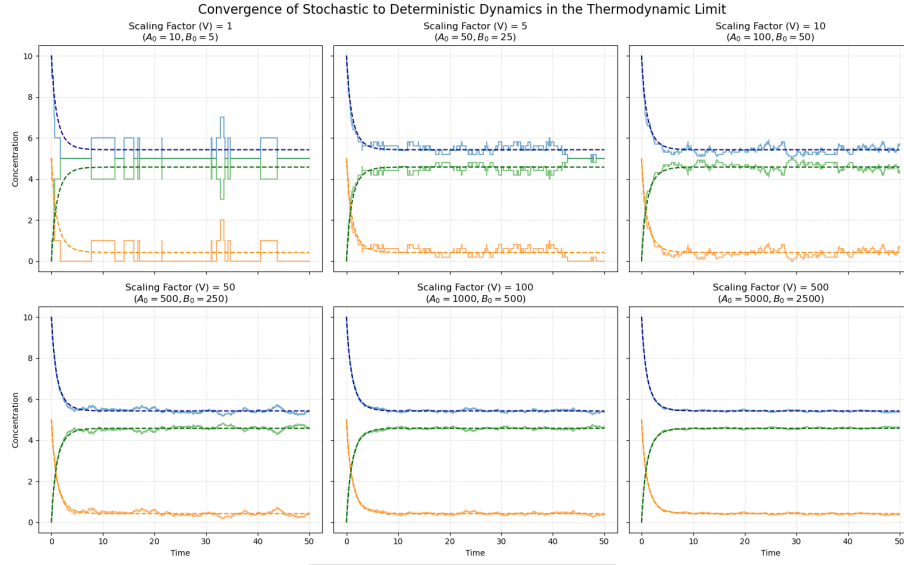


Figure 8: A 2x3 grid comparing stochastic trajectories (solid, semi-transparent lines) to the deterministic ODE solutions (dashed lines) for the reaction $A + B \rightleftharpoons C$. As the system size/volume scaling factor V increases, the stochastic noise diminishes, demonstrating convergence to the macroscopic deterministic limit.

d) Write down the master equation $\pi'(t) = \pi(t) \cdot Q$ for this specific system.

To formulate the Chemical Master Equation (CME) in the matrix form $\pi'(t) = \pi(t)Q$, we first define the state space. Because the reaction $A + B \rightleftharpoons C$ strictly conserves the total mass, the initial quantities $A_{\text{tot}} = n_A + n_C$ and $B_{\text{tot}} = n_B + n_C$ are constant throughout the evolution of the system. Therefore, the entire state can be uniquely described by a single discrete random variable $k = n_C$, which represents the number of C molecules. The state space is bounded such that $k \in \{0, 1, \dots, N\}$, where $N = \min(A_{\text{tot}}, B_{\text{tot}})$.

We define the system's probability state vector $\pi(t)$ as a row vector of length

$N + 1$:

$$\pi(t) = [\pi_0(t), \pi_1(t), \dots, \pi_N(t)]$$

where $\pi_k(t)$ is the probability that exactly k molecules of C exist at time t .

The time evolution of these probabilities is governed by the transition rate matrix (or infinitesimal generator) Q . For any state k , the possible transitions in an infinitesimal time step are limited to the forward reaction (moving to $k + 1$) and the backward reaction (moving to $k - 1$). Using the stochastic rate constants c_1 and c_2 , the nonzero elements of Q are defined as follows:

$$\begin{aligned} Q_{k,k+1} &= c_1(A_{\text{tot}} - k)(B_{\text{tot}} - k) && \text{(Forward reaction rate)} \\ Q_{k,k-1} &= c_2k && \text{(Backward reaction rate)} \\ Q_{k,k} &= -(Q_{k,k+1} + Q_{k,k-1}) && \text{(Probability conservation)} \end{aligned}$$

All other elements in Q (where $|i - j| > 1$) are zero, which structures Q as a tridiagonal matrix.

Expanding the matrix multiplication $\pi'(t) = \pi(t)Q$, the specific differential equation for the probability of being in state k is given by the k -th component:

$$\frac{d\pi_k(t)}{dt} = \pi_{k-1}(t)Q_{k-1,k} + \pi_{k+1}(t)Q_{k+1,k} + \pi_k(t)Q_{k,k}$$

Substituting the specific reaction propensities for this network yields the final, explicit Master Equation for state k :

$$\begin{aligned} \frac{d\pi_k(t)}{dt} &= c_1(A_{\text{tot}} - k + 1)(B_{\text{tot}} - k + 1)\pi_{k-1}(t) \\ &\quad + c_2(k + 1)\pi_{k+1}(t) \\ &\quad - [c_1(A_{\text{tot}} - k)(B_{\text{tot}} - k) + c_2k]\pi_k(t) \end{aligned}$$

This equation explicitly describes how the probability mass flows into state k from the adjacent states $k - 1$ and $k + 1$, while simultaneously flowing out of state k back into those same adjacent states.

Problem 3

Consider the SIR model in reaction form, $S + I \rightarrow 2I$ with rate constant β , and $I \rightarrow R$ with rate constant γ . Here S , I , R stand for susceptible, infected, and recovered individuals.

a) Write down the associated deterministic ODE for this system, and confirm that it is the classic SIR model.

Model Formulation from Reactions

We consider a compartmental epidemiological model represented by the following macroscopic reaction network:



where S , I , and R represent the populations of susceptible, infected, and recovered individuals, respectively.

Applying the principles of mass-action kinetics to this reaction scheme, the rate of the infection reaction is proportional to the product of the interacting populations (βSI), while the rate of the recovery reaction is proportional only to the infected population (γI).

By determining the net rates of creation and consumption for each species, we can derive the associated deterministic Ordinary Differential Equations (ODEs):

$$\begin{aligned} \frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I \end{aligned}$$

Confirmation of the Classic SIR Model

The system of equations derived above perfectly matches the classic SIR model. The first equation shows that susceptible individuals are depleted as they contact infected individuals and contract the disease. The second equation shows that the infected population grows through these new infections (βSI) but decreases as individuals recover (γI). The final equation tracks the growth of the recovered population, which directly mirrors the rate of recovery from the infected compartment.

Furthermore, summing the three differential equations yields:

$$\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = -\beta SI + (\beta SI - \gamma I) + \gamma I = 0$$

This confirms that the total population $N = S(t) + I(t) + R(t)$ remains constant over time, a fundamental assumption of the standard SIR model representing a closed population with no vital dynamics (births or deaths).

b) For initial conditions $S(0) = N$ large, $I(0) = \epsilon$ small, and $R(0) = 0$, show that $I'(0) > 0$ if and only if $\frac{\beta N}{\gamma} > 1$. This constant has the name R_0 or reproductive number.

To determine whether an epidemic will occur when a disease is introduced

into a population, we analyze the initial rate of change of the infected population. We assume the total population is N . Initially, almost everyone is susceptible ($S(0) = N$), the number of initial infections is very small ($I(0) = \epsilon$, where $\epsilon \ll N$), and there are no recovered individuals ($R(0) = 0$).

From the derived ODE system, the rate of change of the infected population is given by:

$$\frac{dI}{dt} = \beta SI - \gamma I = I(\beta S - \gamma)$$

Evaluating this derivative at the initial time $t = 0$:

$$I'(0) = I(0)(\beta S(0) - \gamma) = \epsilon(\beta N - \gamma)$$

For an outbreak to occur, the number of infected individuals must initially increase, meaning $I'(0) > 0$. Because $\epsilon > 0$ (there must be at least some initial infection) and $\gamma > 0$ (individuals take some time to recover), we can isolate the condition for an outbreak:

$$\begin{aligned}\epsilon(\beta N - \gamma) &> 0 \\ \beta N - \gamma &> 0 \\ \beta N &> \gamma \\ \frac{\beta N}{\gamma} &> 1\end{aligned}$$

Thus, the infected population will grow if and only if $\frac{\beta N}{\gamma} > 1$.

This dimensionless quantity is formally defined as the **basic reproduction number**, denoted R_0 :

$$R_0 = \frac{\beta N}{\gamma}$$

Biologically, R_0 represents the average number of secondary infections produced by a single infected individual introduced into a completely susceptible population.

- The term $\frac{1}{\gamma}$ represents the average duration of infectiousness. - The term βN represents the rate at which an infected individual successfully infects others when the entire population N is susceptible.

Therefore, multiplying the infection rate by the duration gives the total expected secondary cases. If $R_0 > 1$, the disease successfully replaces each recovery with more than one new infection, leading to an epidemic. If $R_0 < 1$, each infection produces less than one new infection on average, and the disease will naturally die out without causing a widespread outbreak.

c) Set $N = 100$ and $\epsilon = 3$. Simulate the system stochastically for values of R_0 slightly larger than 1, and show by plotting multiple time courses for each parameter choice how when $R_0 > 1$ an epidemic tends to take place.

To visualize the theoretical threshold behavior, we simulate the stochastic version of the SIR model using the Gillespie algorithm [chart:30]. The total population is fixed at $N = 100$, and the initial infection count is small ($\epsilon = 3$). We simulate the system for three values of the basic reproduction number that are progressively larger than the theoretical threshold of 1: $R_0 = 1.1$, $R_0 = 1.5$, and $R_0 = 2.5$. To achieve this, the recovery rate is fixed ($\gamma = 0.1$), and the infection rate β is adjusted according to the relation $\beta = \frac{R_0 \gamma}{N}$.

For each R_0 value, the model is run five times to capture the inherent stochasticity of the early infection phase. The time courses of the infected population (I) are plotted in Figure 9.

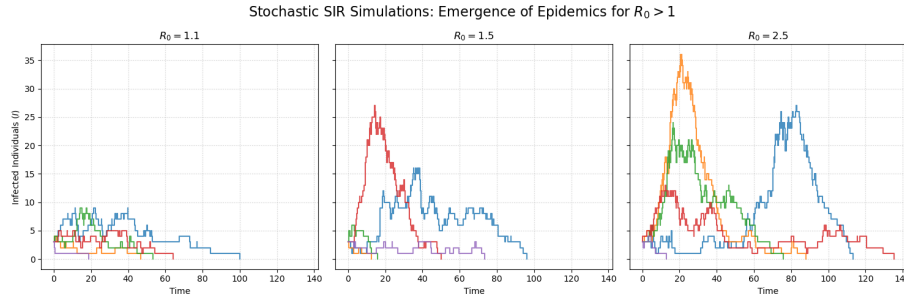


Figure 9: Multiple stochastic realizations of the SIR model for varying $R_0 > 1$ ($N = 100, \epsilon = 3$). As R_0 increases, the probability of a major epidemic increases, the peak infection size is drastically magnified, and the timing of the outbreak is accelerated.

The simulations clearly demonstrate the epidemic-inducing behavior when $R_0 > 1$:

- **At $R_0 = 1.1$:** The parameter is only slightly above the threshold. While the system favors epidemic growth, the weakness of the infection rate combined with demographic stochasticity causes significant variation. Some runs result in minor flare-ups that die out quickly, while others manage to create slow, prolonged, but relatively small epidemics peaking at less than 10 concurrent infections.
- **At $R_0 = 1.5$:** The epidemic behavior becomes more reliable. The curves show a clearer, more consistent exponential growth phase followed by a decline as susceptible individuals are depleted. The peak of the infection is higher (around 15-20 individuals) and occurs earlier.
- **At $R_0 = 2.5$:** The disease is highly infectious. The initial small infection number rapidly explodes, resulting in severe outbreaks that infect a massive proportion of the population concurrently (peaking around 30-40 individuals). The outbreaks happen much faster and are highly determin-

istic in their macroscopic shape, as the strong infection pressure quickly overpowers the early stochastic noise.

These results validate the mathematical condition: as soon as $R_0 > 1$, the disease has the biological capacity to trigger a widespread epidemic, and increasing R_0 exacerbates both the severity and the speed of the outbreak.

d) Set $N = 100$ and $\epsilon = 3$. Simulate the system stochastically for values of R_0 smaller than 1, and show by plotting multiple time courses for each parameter choice how when $R_0 < 1$ an epidemic does not tend to take place.

Conversely, we verify the theoretical prediction for $R_0 < 1$ by simulating the stochastic SIR model with the basic reproduction number set below the epidemic threshold [chart:31]. The initial conditions remain identical ($N = 100, \epsilon = 3$), and the recovery rate is fixed at $\gamma = 0.1$. We test three sub-threshold values: $R_0 = 0.9$, $R_0 = 0.5$, and $R_0 = 0.2$. The resulting time courses for the infected populations are shown in Figure 10.

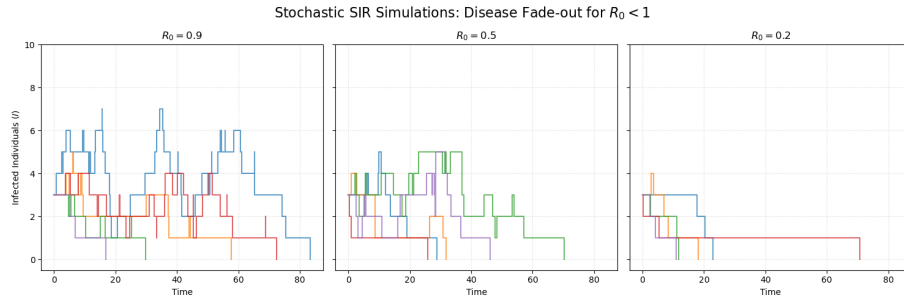


Figure 10: Multiple stochastic realizations of the SIR model for varying $R_0 < 1$ ($N = 100, \epsilon = 3$). The y-axis is scaled down to a maximum of 10. Regardless of the parameter choice, the disease fails to trigger an epidemic and reliably fades out.

The simulations corroborate that an epidemic does not occur when the basic reproduction number is less than 1:

- **At $R_0 = 0.9$:** The system is just below the threshold. While one or two runs show brief, stuttering growth that reaches 5 or 6 infected individuals due to chance (demographic stochasticity), the overall trend is inevitably downward. The infection cannot sustain itself, and the outbreak dies off completely within 60 time units.
- **At $R_0 = 0.5$:** The failure of the disease to spread is much more pronounced. The recovery reaction heavily dominates the infection reaction. Any new secondary infections are quickly offset by multiple recoveries, causing the infected population to rapidly decay toward zero.

- **At $R_0 = 0.2$:** The disease is highly contained. In almost all realizations, the initial 3 infected individuals recover before transmitting the disease to anyone else, resulting in an immediate and near-instantaneous collapse of the outbreak.

Ultimately, these stochastic time courses confirm that when $R_0 < 1$, the introduction of a few infected individuals will only lead to minor, localized stuttering chains of transmission that quickly undergo stochastic extinction, fully preventing the onset of a large-scale epidemic.

e) Argue why R_0 can be considered to be the average number of people infected by each infectious individual. *Hint: Calculate the average time that an infected person remains infected, and with what frequency they infect other individuals during that time.*

The constant $R_0 = \frac{\beta N}{\gamma}$ is mathematically defined as the threshold parameter for an epidemic, but it is more intuitively understood as the average number of secondary infections generated by a single infected individual introduced into a completely susceptible population.

This biological interpretation can be derived by breaking down the infectious process into its two core components: the average duration of the infection and the rate at which the infected individual transmits the disease during that time.

1. Average Duration of Infection

In our SIR model, the recovery of infected individuals follows a first-order reaction $I \xrightarrow{\gamma} R$. Mathematically, this means the time an individual remains in the infectious state (I) before transitioning to the recovered state (R) is an exponentially distributed random variable. The probability density function for the time to recovery τ is $f(\tau) = \gamma e^{-\gamma\tau}$. The expected value, or average time an individual remains infected, is the mean of this exponential distribution:

$$\text{Average Infectious Period} = \int_0^{\infty} \tau \gamma e^{-\gamma\tau} d\tau = \frac{1}{\gamma}$$

2. Frequency of Transmission

Next, we determine how frequently this infectious individual transmits the disease while they are infected. The overall rate of new infections in the population is given by mass-action kinetics as βSI . Therefore, the rate at which a *single* infected individual ($I = 1$) generates new infections is simply βS .

Because we are considering the very beginning of the outbreak, we assume the entire population is susceptible, meaning $S \approx N$. Thus, the frequency at

which our single infected person infects others is:

$$\text{Transmission Rate per Infected Person} = \beta N$$

3. Calculating Total Secondary Infections

To find the total number of people infected by this single individual, we simply multiply the frequency at which they infect others (cases per unit time) by the average duration they remain infectious (total time):

$$R_0 = (\text{Transmission Rate}) \times (\text{Average Infectious Period})$$

$$R_0 = (\beta N) \times \left(\frac{1}{\gamma} \right)$$

$$R_0 = \frac{\beta N}{\gamma}$$

This calculation directly yields our previously derived threshold constant. It perfectly illustrates why an epidemic occurs only if $R_0 > 1$: if a single infectious person causes, on average, more than one new infection before recovering, the number of infected individuals will compound and grow exponentially. If they cause less than one new infection on average ($R_0 < 1$), the transmission chain is unsustainable and the outbreak will inevitably collapse.